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SI and agent tools

An overview: what the tools can do, and what it takes to make it hold

In brief. AI agents are everywhere. The tools to build them are many, and they look alike. The hard part isn't building an agent. The hard part is making it **hold** when it meets your data and your systems. Here's the overview — and what makes the difference.

01 WHAT AN AI AGENT IS

An AI agent is more than a model that answers. It acts. It fetches data, uses tools and takes several steps toward a goal. An agent tool — a toolkit or a toolbox — is the frame you build and steer the agent in.

02 THE TOOLS — WHAT THEY ARE, AND WHAT THEY CAN DO

Most tools do one or two things in the chain: they steer the agent's steps, fetch knowledge, or connect it to your systems. Here are the main ones, briefly:

LangChain / LangGraph	Steers the agent's steps and memory. Most widely used.
Microsoft Agent Framework	Microsoft's stack (AutoGen + Semantic Kernel). For Azure and .NET.
CrewAI	Several agents in roles, collaborating on a task.
OpenAI Agents SDK	An easy path to agents on top of OpenAI. Replaces Swarm.
Google ADK + A2A	Google's kit, plus a protocol so agents can talk to each other.
LlamaIndex	Pulls knowledge into the agent. For data- and document-heavy tasks.
Hugging Face smolagents	Lightweight and open. For small, self-hosted setups.
Anthropic MCP	Not a frame, but the standard that connects agent to data and tools.
NVIDIA Agent Toolkit	Compute, models and runtime. The foundation it all runs on.

03 THE MATURITY STAIRCASE — THROUGH LOOPLO GLASSES

The market draws maturity as a staircase: six steps, each unlocking more. The staircase is fine. But every step rests on a coupling — and it's **the coupling**, not the step, that decides whether it **hold**s.

STEP	WHAT THE MARKET SEES	WHAT WE SEE — THE COUPLING SOMEONE MUST TAKE CARE OF
1 · Basic LLM	Text generation	Model ↔ your language: it knows the language, not your concepts
2 · Long context	Reads your files	Document ↔ meaning: it reads, but doesn't know what a case is at your place
3 · RAG + tools	Fresh data + actions	Retrieval ↔ source: which source, which version, who owns it
4 · Multimodal	Sees and hears	Input ↔ boundary: what may live where, and who has access
5 · Autonomous agent	Autonomy	Action ↔ accountability: what it may do, and who answers
6 · Multi-agent	Collaboration at scale	Agent ↔ agent: the arrows between them are where it breaks

The higher you climb, the more couplings. Without an owner, scale just multiplies the debt. That's why Looplo isn't a step on the staircase — we're the ground it stands on.

04 WHAT THEY HAVE IN COMMON

Step back, and the list is remarkably alike. It's all build-tools. They help you build something new, fast. But none of them tell you how the parts hang together — or what happens when the agent meets your real data and systems.

05 WHERE IT GOES WRONG

An agent project rarely breaks in the model itself. It breaks in **the coupling**: the interplay between agent, data and system. No one owns that coupling, so it sits loose. It works in a demo. It falls apart in production.

It's not a rare fault:

⅓+

More than one in three AI projects is expected to be shut down by 2027.

— GARTNER

91%

already run agentic AI — plenty of projects to lose.

— EY

06 WHAT SI ADDS

SI is **semantic intelligence**: the agent understands your concepts and your context, not just the text. Instead of stacking agents and tools on top of each other and hoping, you ground them in **a model you own** — your data, your rules, and how it all hangs together. Then **the coupling** is no longer something **no one owns**. It's modeled. And then it **hold**s — even as you grow.

OVERVIEW AND INSIGHT

An overview is knowing the tools. Insight is knowing where it breaks — and taking care of **the coupling**. That's the difference between **a demo and something that runs every day**.